Research Methodology

Week 3 Assignment: Questions, Answers and Explanations

Week 3: Data Analysis, Modelling Skills and Laboratory Safety

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. This week covers lab safety, signals, estimation and models.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Select all that apply	**A, B, C, D**	1 (full marks)
2	Single answer	**C**	1 (full marks)
3	Single answer	**B**	1 (full marks)
4	Single answer	**B**	1 (full marks)
5	Select all that apply	**A, C, D**	0.33 (partial: only one of the three incorrect statements chosen)
6	Single answer	**C**	1 (full marks)
7	Single answer	**C**	1 (full marks)
8	Single answer	**D**	0 (wrong answer chosen)
9	Select all that apply	**A, C**	0.5 (partial: only one of the two correct options chosen)
10	Single answer	**C**	1 (full marks)

Questions with explanations

Question 1 of 10

Safety must be the most important discipline of our routine when we are in laboratory or industrial environment. The common sources which have potential to cause damage, available in our lab towards which we need to be careful are-

Select all that apply.

Option	Statement	Verdict	Reason
A	**Sources of heat**	**Correct**	Heat causes burns and fires.
B	**Chemicals (Acids, Bases etc.)**	**Correct**	They corrode, burn and poison.
C	**Pressurized gas cylinders**	**Correct**	They can leak or burst under pressure.
D	**Electricity and water source**	**Correct**	Electricity together with water causes shocks and short circuits.

Correct answer: A and B and C and D.

Explanation: All four are standard laboratory hazards, so all four need care.

Question 2 of 10

Which of the following statistics is used to quantify errors in parameter estimates?

Single answer.

Option	Statement	Verdict	Reason
A	Mode	**Wrong**	The mode is the most frequent value; it describes the centre of data, not error.
B	Skewness	**Wrong**	Skewness describes the asymmetry of a distribution.
C	**Standard deviation**	**Correct**	The standard deviation (standard error) of an estimate shows how far it typically falls from the true value.
D	Median	**Wrong**	The median is the middle value, again a measure of centre.

Correct answer: C.

Explanation: The error in a parameter estimate is measured by its spread, that is, its standard deviation (called the standard error). Mode, median and skewness describe where the data lie or how they are shaped.

Question 3 of 10

A temperature sensor is installed at the bottom of the cooling tower storage tank to measure the temperature of outlet water. The sensor did not function properly for some time and gave data with higher temperature values. The conclusion we can draw from the process is-

Single answer.

Option	Statement	Verdict	Reason
A	Since data with different values were recorded, this increased the possibility of wide analysis thus very useful for our work	Wrong	Wrong values do not add useful variety; they mislead the analysis.
B	**The data obtained cannot be reliable because these are not true data from the system but the malfunctioned one**	Correct	The readings reflect the faulty sensor, not the process.
C	The data can be used as with good accuracy using the predictive type data analysis where we can assume the extent of error and finally subtract the error from final value	Wrong	The size of a sensor fault is unknown, so it cannot be assumed and subtracted.
D	None of these	Wrong	Option B is a correct conclusion.

Correct answer: B.

Explanation: Data are only as good as the instrument that produced them. A sensor that malfunctioned did not record the true state of the system, so the data from that period cannot be trusted.

Question 4 of 10

Which of the following can be treated as deterministic signals?

Single answer.

Option	Statement	Verdict	Reason
A	Measurements of atmospheric temperature	Wrong	Real measurements contain random variation, so they are not deterministic.
B	**A sine wave corrupted by a rectangular pulse**	Correct	Both parts are fixed mathematical functions of time, so the whole signal is fully predictable.
C	Estimates of signals from measurements	Wrong	Estimates computed from noisy data are themselves random.
D	None of the above	Wrong	Option B qualifies.

Correct answer: B.

Explanation: A deterministic signal is completely described by a mathematical rule with no randomness. A sine wave plus a rectangular pulse follows such a rule, whereas measured or estimated quantities carry random errors.

Question 5 of 10

Identify the incorrect statement(s) among the following w.r.t. non-linear models.

Select all that apply. *Your portal score: 0.33 (partial: only one of the three incorrect statements chosen).*

Option	Statement	Verdict	Reason
A	**Linearization involves mere discarding of the non-linear terms in a model.**	Correct	This statement is incorrect: linearization is a Taylor series expansion about a nominal point keeping the first-order terms, not just dropping terms.
B	The standard Taylor's series-based linearized model changes with the nominal point of linearization.	Wrong	This statement is true: the slope, and so the linear model, depends on the chosen point.
C	**A non-linear model can never yield a zero-valued output to a non-zero input.**	Correct	This statement is incorrect: for example sin x is 0 at x = π, a non-zero input.
D	**Estimates of linear models have smaller errors than those of non-linear models.**	Correct	This statement is incorrect: there is no such rule; error depends on the data and how well the model fits.

Correct answer: A and C and D.

Explanation: Three statements are false. Linearization is a systematic expansion around a nominal operating point, a non-linear model can output zero for a non-zero input, and linear models do not automatically give smaller estimation errors. Only the second statement is true.

Note: *This question asks for the INCORRECT statements, so "Correct" in the verdict column marks the statements you should select.*

Question 6 of 10

Which of the following transforms are used to detect periodic (oscillatory) components in a signal?

Single answer.

Option	Statement	Verdict	Reason
A	Hilbert transform	Wrong	It gives amplitude and phase information, not the set of frequencies present.
B	z-transform	Wrong	It is mainly a tool for analysing discrete-time systems.
C	**Fourier transform**	Correct	It breaks a signal into sine waves of different frequencies, so peaks reveal periodic components.
D	All of the above	Wrong	Only the Fourier transform is the standard tool for this purpose.

Correct answer: C.

Explanation: The Fourier transform expresses a signal as a sum of sinusoids. A strong peak at some frequency tells you the signal has an oscillation at that frequency.

Question 7 of 10

The ratio of errors in parameter estimates of a model from two data sets with SNR 9 and 100 is

Single answer.

Option	Statement	Verdict	Reason
A	3 : 10	**Wrong**	This is the inverted ratio: the noisier data set has the larger error.
B	5 : 1	**Wrong**	It does not follow from the square roots of 9 and 100.
C	**10 : 3**	**Correct**	Error is proportional to 1/√SNR: (1/3) : (1/10) = 10 : 3.
D	10 : 0.9	**Wrong**	It does not follow from taking the square roots of 9 and 100.

Correct answer: C.

Explanation: SNR (signal-to-noise ratio) is signal power divided by noise power, and the error in an estimate grows with the noise level, which is proportional to 1/√SNR. For SNR 9 the error is proportional to 1/√9 = 1/3, and for SNR 100 it is proportional to 1/√100 = 1/10. The ratio (1/3) : (1/10) equals 10 : 3, so the cleaner data set has the smaller error.

Question 8 of 10

Which of the following is true concerning a parameter estimate?

Single answer. *Your portal score: 0 (wrong answer chosen).*

Option	Statement	Verdict	Reason
A	It is in general close to the true value with high probability.	**Wrong**	Closeness depends on the noise and the method; it is not guaranteed in general.
B	When averaged across a million experimental records under same operating conditions, the estimate equals the true value.	**Wrong**	Averaging gives the true value only for an unbiased estimator.
C	Least squares method always gives the estimate with lowest error.	**Wrong**	Least squares is not always the lowest-error method; it depends on the noise.
D	**It is a mathematical function of the observations.**	**Correct**	An estimate is computed from the data by a formula, so it changes with the data.

Correct answer: D.

Explanation: An estimate is a rule applied to the observed data. Because the data are random, the estimate is random too, and the other statements claim guarantees that the definition does not give.

Question 9 of 10

Which of the following statements is true?

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	**Deterministic phenomena produce outcomes that can be predicted accurately.**	**Correct**	Deterministic means the outcome follows from fixed rules.
B	Random processes occur randomly.	**Wrong**	This is circular; random processes follow probability laws.
C	**A process can produce a mix of deterministic and stochastic effects.**	**Correct**	Real processes usually have a predictable part plus random noise.
D	Signals that are random can never take negative values.	**Wrong**	Random signals such as noise can be negative.

Correct answer: A and C.

Explanation: Deterministic outcomes can be predicted, and real processes mix predictable and random parts. Randomness has nothing to do with sign, and "random processes occur randomly" says nothing useful.

Question 10 of 10

An estimator is said to be most efficient when

Single answer.

Option	Statement	Verdict	Reason
A	The parameters can be estimated in the least possible time.	**Wrong**	Computation time is a separate property from statistical efficiency.
B	The estimator provides accurate estimates with the smallest number of observations.	**Wrong**	This describes sample size, not the formal definition used in the answer key.
C	**Standard error in the parameter estimates is lowest when compared to that of all other estimators.**	**Correct**	Efficiency compares estimators: the most efficient has the smallest standard error.
D	It estimates parameters accurately even in presence of outliers.	**Wrong**	That property is robustness, not efficiency.

Correct answer: C.

Explanation: In statistics, efficiency compares estimators on the same data. The most efficient estimator has the lowest standard error (variance) among all estimators.